

Healthcare Analytics Dashboard – Power BI Project

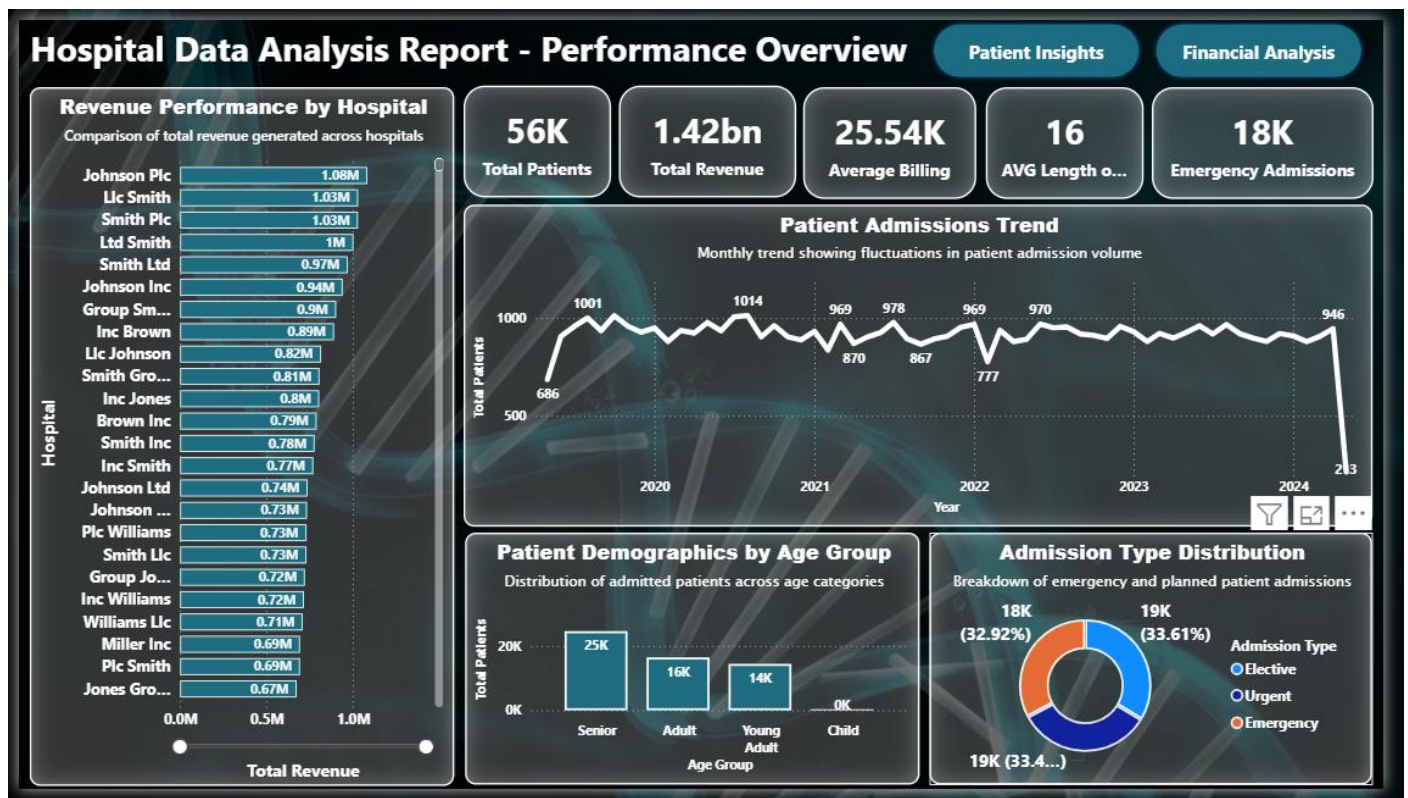
Project Introduction

Healthcare institutions generate large volumes of operational and clinical data every day. However, without proper analysis, this data cannot support effective decision-making. This project focuses on transforming raw hospital patient data into meaningful business insights using **Power BI**.

The objective of this project was to design interactive dashboards that provide clear visibility into **hospital performance, patient health trends, and financial outcomes**. Each dashboard was built with the goal of helping healthcare stakeholders make faster and smarter decisions based on real data patterns.

Dashboard 1: Hospital Performance Overview

This dashboard provides a high-level understanding of hospital operations, patient flow, and overall performance.



KPI Cards:

Total Patients

Displays the total number of patients admitted within the selected timeframe. This metric helps evaluate hospital demand and operational workload.

Total Revenue

Represents the overall billing revenue generated from patient services. It reflects the financial performance of hospital operations.

Average Billing Amount

Shows the average cost incurred per patient admission. This helps assess pricing patterns and cost consistency across treatments.

Average Length of Stay

Indicates the average number of days patients remain admitted. This metric is useful for evaluating treatment efficiency and hospital resource utilization.

Emergency Admissions

Highlights the number of patients admitted through emergency services, helping understand urgent care demand.

Visualizations:

Patient Admissions Trend

This visual illustrates how patient admissions fluctuate over time. It helps identify seasonal demand, peak admission periods, and overall growth trends.

Revenue by Hospital

This comparison chart highlights revenue generated by each hospital. It helps identify high-performing facilities and those requiring operational improvements.

Admission Type Distribution

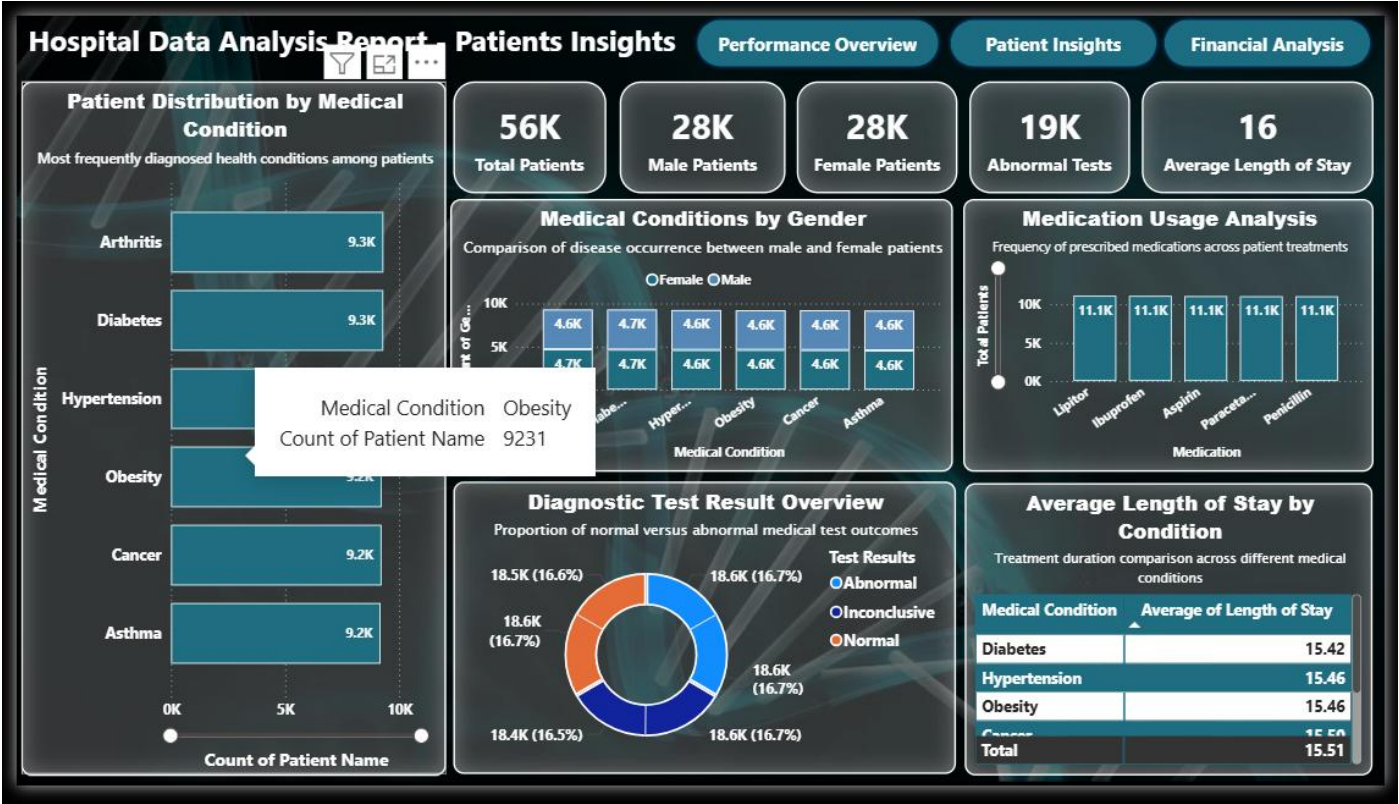
Displays the proportion of emergency and planned admissions. This helps hospitals understand patient inflow patterns and operational planning requirements.

Patient Demographics by Age Group

Shows patient distribution across different age categories, helping hospitals understand which demographic segments require the most medical attention.

Dashboard 2: Clinical & Patient Insights

This dashboard focuses on patient health conditions, treatment complexity, and clinical performance.



KPI Cards:

Male Patients

Displays the total number of male patients receiving treatment, supporting gender-based health analysis.

Female Patients

Shows the total number of female patients treated, enabling comparative healthcare utilization analysis.

Abnormal Test Results

Highlights patients with abnormal diagnostic outcomes, helping clinicians assess patient risk levels and diagnostic workload.

Average Length of Stay

Measures treatment duration across medical conditions, providing insights into recovery timelines.

Visualizations:

Patient Distribution by Medical Condition

Identifies the most frequently treated diseases, helping hospitals allocate medical resources effectively.

Medical Conditions by Gender

Compares disease occurrence between male and female patients, supporting gender-focused healthcare strategies.

Diagnostic Test Result Overview

Shows the ratio of normal and abnormal test results, providing a quick summary of overall patient health outcomes.

Medication Usage Analysis

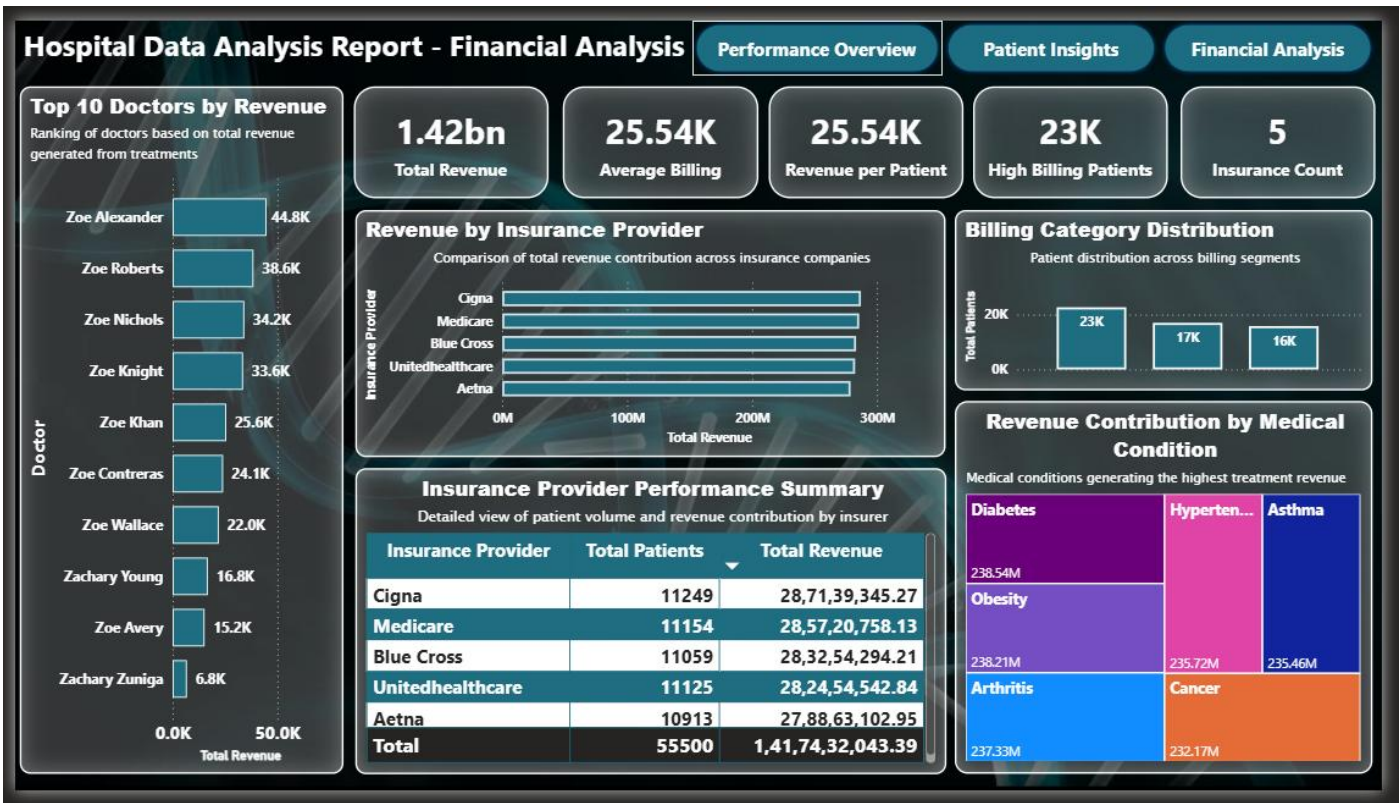
Highlights commonly prescribed medications, supporting inventory management and treatment planning.

Average Length of Stay by Condition

Displays treatment duration across different diseases, helping evaluate medical complexity and patient recovery time.

Dashboard 3: Financial & Insurance Analysis

This dashboard focuses on financial performance, billing trends, and insurance provider contributions.



KPI Cards:

Total Revenue

Represents the overall income generated from patient treatments and services.

Average Billing Amount

Shows the average amount billed per patient, helping assess pricing efficiency.

Revenue per Patient

Measures profitability by calculating the average revenue generated from each patient admission.

High Billing Patients

Identifies patients with significantly higher treatment costs, helping hospitals monitor expensive medical cases.

Visualizations:

Insurance Provider Coverage

Displays the number of insurance providers contributing to hospital billing, reflecting coverage diversity.

Revenue by Insurance Provider

Compares revenue generated from different insurance companies, helping identify valuable partnerships.

Revenue Contribution by Medical Condition

Highlights which medical conditions generate the highest treatment revenue, helping hospitals understand cost drivers.

Billing Category Distribution

Shows how patients are distributed across low, medium, and high billing categories, helping evaluate overall billing structure.

Top Doctors by Revenue Generation

Ranks doctors based on revenue generated from patient treatments, recognizing high-impact contributors.

Insurance Provider Performance Summary

Provides a detailed comparison of patient volume and revenue contribution across insurance providers.

Key Analytical Enhancements

To strengthen the analysis, several calculated metrics were created, including:

- Patient volume and revenue aggregation measures
- Emergency admission tracking
- Length of stay calculations
- Revenue per patient profitability metric
- Patient segmentation using age and billing categories

These calculations enhanced the dashboard's ability to deliver deeper insights beyond basic reporting.

Challenges Faced During Development

Data Standardization

Inconsistent naming formats across hospitals and doctors created duplicate entries. This was resolved using transformation techniques in Power Query.

Meaningful KPI Selection

Raw numerical metrics initially lacked business relevance. Additional derived KPIs were created to provide actionable insights.

Visual Selection & Layout Design

Selecting appropriate visualizations required multiple iterations to ensure clarity, readability, and storytelling effectiveness.

Data Interpretation Balance

Maintaining the balance between detailed analysis and dashboard simplicity was a key design challenge. KPIs and summary visuals were structured to ensure user-friendly navigation.

Tools & Technologies Used

- Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Data Modelling Techniques
- Visualization & Dashboard Design Principles

Final Project Summary

This project demonstrates the practical application of business intelligence techniques in the healthcare domain. By transforming raw patient and hospital data into interactive dashboards, the project provides meaningful insights into operational efficiency, clinical trends, and financial performance.

The dashboards are designed to support data-driven decision-making while maintaining clarity and accessibility for non-technical stakeholders. This project reflects strong capabilities in data preparation, visualization design, analytical thinking, and business insight generation.

“ This project represents my commitment to building analytics solutions that not only present data but also deliver actionable value.”

“Thank You”